Model-based Nonlinear Control Applications with Kinematic and Dynamic Analysis of 3 DoF RRR Planar Robot Manipulator

Şaheste Berfin Gençtor[a], Halil Uzunoğlu[b,*]

[a] Department of Electrical and Electronics Engineering, İzmir Katip Çelebi University, İzmir, Türkiye

Abstract

This study presents a model-based nonlinear control framework developed for a 3 DoF RRR planar robotic manipulator. The full kinematic and dynamic models of the robot are derived and used to design and implement three different control strategies: joint space position control, Cartesian space position control, and hybrid position/force control in Cartesian space. Each controller is evaluated in MATLAB Simulink through simulations performed under predefined reference trajectories and force interaction scenarios.

In the joint space control strategy, dynamic compensation is applied to improve trajectory tracking accuracy by eliminating the effects of nonlinearities and joint couplings. The Cartesian space controller enables direct end-effector control, resulting in high-precision motion along complex planar paths. The hybrid controller combines position regulation in the x and y directions with force control in the z direction, allowing the end-effector to interact compliantly with a simulated surface environment.

Simulation results confirm that all three controllers achieve stable and accurate performance. The hybrid controller, in particular, demonstrates the ability to manage motion and contact force simultaneously, making it suitable for applications involving interaction, such as insertion, polishing, or surface tracking. The proposed framework highlights the effectiveness of model-based nonlinear control in enhancing the accuracy, responsiveness, and adaptability of planar robotic systems.

Keywords: *Cartesian space; Dynamic model; Joint space; Kinematic model; Nonlinear control; Robotic arm.*

1. Introduction

In recent years, robotic arms have become increasingly indispensable in both industrial and research applications due to their versatility, repeatability, and potential for automation. With the growing demand for precision tasks in planar workspaces planar robotic manipulators have gained significant attention. Compared to traditional serial robots with higher degrees of freedom, planar manipulators offer a simpler structure, more efficient control schemes, and sufficient performance for many two-dimensional tasks. Among these, the 3 Degree-of-Freedom (DoF) RRR planar manipulator has emerged as a widely studied model due to its ability to perform fully-actuated movements in the Cartesian x-y plane.

The 3 DoF RRR manipulator, consisting entirely of revolute joints, enables smooth and continuous motion in planar spaces and provides a compact structure ideal for high-speed and precise positioning tasks. Because of its reduced complexity and direct geometric interpretation, this structure is frequently used as a benchmark for evaluating advanced control algorithms in both academic and simulation-based environments.

Several model-based control strategies have been proposed in the literature to enhance the accuracy and robustness of robotic manipulators. In particular, nonlinear control techniques that incorporate full dynamic models have shown promising results in achieving high-fidelity trajectory tracking and dynamic compensation [1]. However, while many studies have addressed joint space control, the extension of these methods to Cartesian space—especially to hybrid position/force control—remains an active research area, especially in planar systems.

This study aims to address this gap by developing and comparing three model-based nonlinear controllers for a 3 DoF RRR planar robotic arm: joint space position control, Cartesian space position control, and hybrid position/force control in Cartesian space. Each control method is implemented and validated in MATLAB Simulink, using a full kinematic and dynamic model of the manipulator. The goal is to investigate the performance of these control architectures in terms of trajectory tracking accuracy, force regulation capability, and compliance with environmental interaction—factors that are essential in modern robotic tasks requiring precision and adaptability.

2. Mathematical model of 3 DoF RRR Planar Robot Manipulator

In this section, the kinematic and dynamic model equations of 3 DoF RRR Planar Robot Manipulator are given in detail. As shown in Figure 1, the 3 DoF RRR Planar Robot Manipulator's trajectory generation, forward kinematic, Jacobian, and dynamic model blocks are created in the MATLAB Simulink.

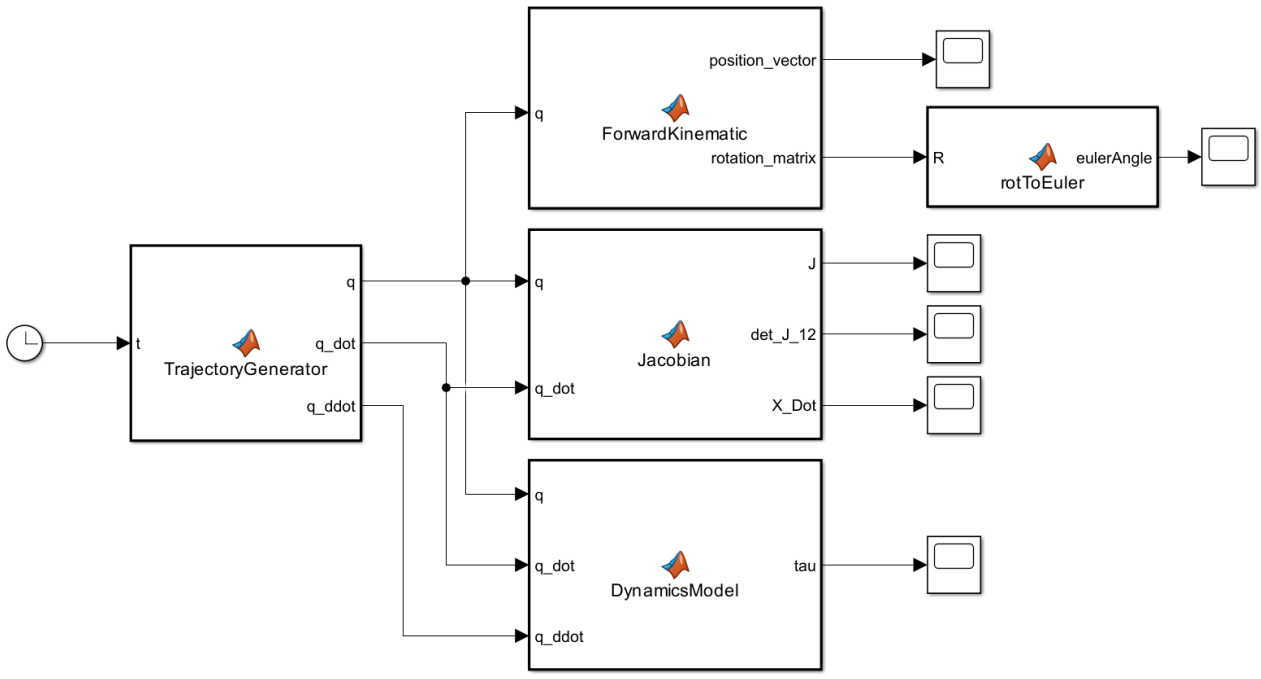


Figure 1: The 3 DoF RRR Planar Robot Manipulator robotic arm's model blocks in MATLAB Simulink.

2.1. Forward Kinematics

Forward kinematics formulates the end-effector position and orientation in terms of joint angles for serial robotic manipulators. The 3 DoF RRR planar manipulator considered in this study lies entirely in a 2D plane, with all revolute joints affecting motion in the x-y plane. Its geometric structure and joint configuration are illustrated in Figure 2, where each link and joint is clearly represented.

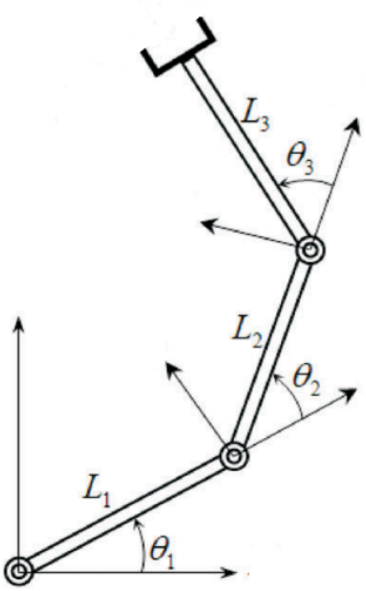


Figure 2: 3 DoF RRR Planar Robot Manipulator [2]

The mathematical model of the manipulator is constructed using the modified Denavit–Hartenberg convention. The relevant kinematic parameters such as link lengths a_i, joint angles θ_i and offsets d_i are tabulated in Table 1, which summarizes the physical parameters required for defining link-to-link transformations.

Table 1: DH Parameters

Link i	θ_i	d_i	a_i	α_i
1	θ_1	0	l_1	0
2	θ_2	0	l_2	0
3	θ_3	0	l_3	0

The general form of the homogeneous transformation matrix between two successive links is given in (1), where θ_i is the joint variable and a_i the corresponding link length:

$$T_i^{i-1} = \begin{bmatrix} cos\theta_i & -sin\theta_i & 0 & a_i cos\theta_i \\ sin\theta_i & cos\theta_i & 0 & a_i sin\theta_i \\ 0 & 0 & 1 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

Using this expression, the individual transformation matrices for the three joints of the manipulator are derived. The first transformation matrix describes a rotation by θ_i followed by a translation along l_1 (2).

$$T_1 = \begin{bmatrix} cos\theta_1 & -sin\theta_1 & 0 & l_1 cos\theta_1 \\ sin\theta_1 & cos\theta_1 & 0 & l_1 sin\theta_1 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{2}$$

The second transformation matrix captures the motion due to the second joint and link l_2, with a similar rotation–translation pattern (3).

$$T_2 = \begin{bmatrix} cos\theta_2 & -sin\theta_2 & 0 & l_2 cos\theta_2 \\ sin\theta_2 & cos\theta_2 & 0 & l_2 sin\theta_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

Likewise, the third link's transformation is modeled in (4), completing the joint chain of the manipulator.

$$T_3 = \begin{bmatrix} cos\theta_3 & -sin\theta_3 & 0 & l_3 cos\theta_3 \\ sin\theta_3 & cos\theta_3 & 0 & l_3 sin\theta_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{4}$$

The overall transformation from the base to the end-effector is obtained by the product of the three individual matrices (5).

$$T_3^0 = T_1 T_2 T_3 \tag{5}$$

This results in a complete expression that reflects both the rotation and translation accumulated across all joints (6).

$$T_3^0 = \begin{bmatrix} cos(\theta_1+\theta_2+\theta_3) & -sin(\theta_1+\theta_2+\theta_3) & 0 & l_1 cos\theta_1 + l_2 cos(\theta_1+\theta_2) + l_3 cos(\theta_1+\theta_2++\theta_3) \\ sin(\theta_1+\theta_2+\theta_3) & cos(\theta_1+\theta_2+\theta_3) & 0 & l_1 sin\theta_1 + l_2 sin(\theta_1+\theta_2) + l_3 sin(\theta_1+\theta_2++\theta_3) \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{6}$$

The rotation matrix portion of T_3^0, showing the orientation of the end-effector, is isolated in (7).

$$R_3^0(\theta) = \begin{bmatrix} cos(\theta_1+\theta_2+\theta_3) & -sin(\theta_1+\theta_2+\theta_3) & 0 \\ sin(\theta_1+\theta_2+\theta_3) & cos(\theta_1+\theta_2+\theta_3) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{7}$$

Finally, the Cartesian position of the end-effector is extracted as a vector $P_3^0(\theta)$, depending explicitly on the joint angles and link lengths (8).

$$P_3^0(\theta) = \begin{bmatrix} l_1 cos\theta_1 + l_2 cos(\theta_1 + \theta_2) + l_3 cos(\theta_1 + \theta_2 + +\theta_3) \\ l_2 sin(\theta_1 + \theta_2) + l_3 sin(\theta_1 + \theta_2 + +\theta_3) \\ 0 \end{bmatrix} \tag{8}$$

2.2. Velocity Kinematics

Velocity kinematics of a serial manipulator describes the relationship between the joint angular velocities and the resulting end-effector linear and angular velocities. For a 3 DoF RRR planar manipulator, the velocities propagate through the kinematic chain, and their analytical representation is crucial for both control and dynamic modeling. The cumulative joint angles are first defined to simplify notation for the transformation derivatives. These include the individual joint angle θ_1, the sum of the first two angles $\theta_{12} = \theta_1 + \theta_2$ and the total sum $\theta_{123} = \theta_1 + \theta_2 + \theta_3$ (9). This representation allows the velocity Jacobian to be expressed compactly using trigonometric terms.

$$\begin{bmatrix} \theta_1 \\ \theta_{12} \\ \theta_{123} \end{bmatrix} = \begin{bmatrix} \theta_1 \\ \theta_1 + \theta_2 \\ \theta_1 + \theta_2 + \theta_3 \end{bmatrix} \tag{9}$$

The spatial velocity of the end-effector frame (frame 3) with respect to the base frame (frame 0), denoted as v_3^0 , comprises the linear velocity $\dot{P}_3^0$ and the angular velocity w_3^0 . These are derived through the time derivative of the forward kinematics transformation, yielding a linear combination of joint velocities weighted by trigonometric terms and link lengths (10). This expression effectively captures how each joint motion influences the Cartesian motion of the end-effector.

$$v_3^0 = \begin{bmatrix} \dot{P}_3^0 \\ \omega_3^0 \end{bmatrix} = \begin{bmatrix} -l_1 sin\theta_1 - l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_3 sin\theta_{123} \\ l_1 cos\theta_1 + l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_3 cos\theta_{123} \\ 1 & 1 & 1 \end{bmatrix} * \begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \end{bmatrix} \tag{10}$$

This velocity relation can be expressed compactly using the Jacobian matrix $J(\theta)$, which maps joint velocities to the Cartesian velocity of the end-effector (11). This linear transformation plays a central role in motion control and dynamic formulation.

$$\dot{P_3^0} = J(\theta)\dot{\theta} \tag{11}$$

The Jacobian matrix is composed of two components: the translational part $J_P(\theta)$, and the rotational part $J_O(\theta)$. The complete Jacobian is shown in (12), with rows corresponding to the contributions of each joint to the spatial motion of the end-effector.

$$J(\theta) = \begin{bmatrix} J_P(\theta) \\ J_O(\theta) \end{bmatrix} = \begin{bmatrix} -l_1 sin\theta_1 - l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_3 sin\theta_{123} \\ l_1 cos\theta_1 + l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_3 cos\theta_{123} \\ 1 & 1 & 1 \end{bmatrix} \tag{12}$$

The translational Jacobian $J_P(\theta)$ is specifically responsible for determining the linear velocity

component of the end-effector (13). Each column corresponds to the contribution of a specific joint to the overall translational velocity.

$$J_P(\theta) = \begin{bmatrix} -l_1 sin\theta_1 - l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_2 sin\theta_{12} - l_3 sin\theta_{123} & -l_3 sin\theta_{123} \\ l_1 cos\theta_1 + l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_2 cos\theta_{12} + l_3 cos\theta_{123} & +l_3 cos\theta_{123} \end{bmatrix} \tag{13}$$

The rotational Jacobian $J_O(\theta)$ is trivial in this planar case, as each joint contributes equally to the rotation about the out-of-plane z-axis (14).

$$J_O(\theta) = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \tag{14}$$

2.3. Dynamic Model

The dynamic behavior of the 3 DoF RRR planar manipulator is derived using the Euler–Lagrange formulation. This model captures how joint torques τ generate motion through joint accelerations and velocity-dependent forces. The general form of the manipulator dynamics is given in (15), where $M(\theta)$ is the symmetric positive definite inertia matrix, and $C(\theta, \dot{\theta})\dot{\theta}$ represents Coriolis and centrifugal forces.

$$\tau = M(\theta)\ddot{\theta} + C(\theta, \dot{\theta})\dot{\theta} \tag{15}$$

The inertia matrix $M(\theta)$, which relates joint accelerations to torques, is a 3×3 symmetric matrix. Its general symbolic structure is defined in (16), and it accounts for the mass and geometry of each link.

$$M(\theta) = \begin{bmatrix} M_{11} & M_{12} & M_{13} \\ M_{12} & M_{22} & M_{23} \\ M_{13} & M_{23} & M_{33} \end{bmatrix} \tag{16}$$

Similarly, the Coriolis and centrifugal matrix $C(\theta, \dot{\theta})$ contains nonlinear velocity-dependent terms that arise due to motion coupling between joints, and is defined in (17).

$$C(\theta, \dot{\theta}) = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} \tag{17}$$

To express the matrices in a compact form, the following notations are used: $c_i = \cos(\theta_i)$, $s_i = \sin(\theta_i)$, $c_{ij} = \cos(\theta_i + \theta_j)$, $s_{ij} = \sin(\theta_i + \theta_j)$ and $c_{ijk} = \cos(\theta_i + \theta_j + \theta_k)$. These terms simplify expressions involving composite joint angles.

- $c_i = cos(\theta_i)$
- $s_i = sin(\theta_i)$
- $c_i j = cos(\theta_i + \theta_j)$

- $s_ij = sin(\theta_i + \theta_j)$
- $c_ijk = cos(\theta_i + \theta_j + \theta_k)$
- $s_ijk = sin(\theta_i + \theta_j + \theta_k)$

The full expression of the inertia matrix in terms of system parameters is provided in (18). Each element is written as a combination of known constants p_i , which represent moments of inertia and mass contributions derived from link geometry (Table 2).

$$M(\theta) = \begin{bmatrix} p_1c_2 + p_2(c_3 + c_{23}) + p_3 & p_6c_2 + p_2c_3 + p_7c_{23} + p_4 & p_7(c_3 + c_{23}) \\ p_6c_2 + p_2c_3 + p_7c_{23} + p_4 & p_2c_3 + p_4 & p_7c_3 + p_5 \\ p_7(c_3 + c_{23}) & p_7c_3 + p_5 & p_5 \end{bmatrix} \tag{18}$$

The Coriolis and centrifugal matrix is derived based on the Christoffel symbols, and its full form is shown in (19). Each element is a product of p_i constants and sine terms multiplied by joint velocities. The structure reflects the inter-joint coupling during acceleration and deceleration phases.

$$C(\theta, \dot{\theta}) = \begin{bmatrix} -(p_1s_2 + p_2s_{23})\dot{\theta}_2 - p_2(s_3 + s_{23})\dot{\theta}_3 & -(p_6s_2 + p_7s_{23})\dot{\theta}_2 - p_2(s_3 + s_{23})\dot{\theta}_3 & -p_7(s_3 + s_{23})\dot{\theta}_3 \\ (p_6s_2 + p_7s_{23})\dot{\theta}_1 & -p_2s_3\dot{\theta}_3 & -(p_2s_3\dot{\theta}_1 + p_7s_3\dot{\theta}_3) \\ p_7(s_3 + s_{23})\dot{\theta}_1 & p_2s_3\dot{\theta}_1 + p_7s_3\dot{\theta}_2 & 0 \end{bmatrix} \tag{19}$$

The constants p_1 to p_7 in these matrices represent physical properties of the links, including mass, center of mass, and moment of inertia. These are listed with their numerical values as follows:

- $p_1 = 21.3 \times 10^{-3} Kg.m^2$
- $p_2 = 2.9 \times 10^{-3} Kg.m^2$
- $p_3 = 43.3 \times 10^{-3} Kg.m^2$
- $p_4 = 17.7 \times 10^{-3} Kg.m^2$
- $p_5 = 1.7 \times 10^{-3} Kg.m^2$
- $p_6 = 10.6 \times 10^{-3} Kg.m^2$
- $p_7 = 1.5 \times 10^{-3} Kg.m^2$

These constants are calculated from the physical structure of the robot, and are derived from the link parameters given in Table 2, which includes link lengths, masses, center of mass locations, and principal inertias.

Table 2: Inertia Parameters for Three Links

Link	Link 1	Link 2	Link 3
Length (m)	l_1	l_2	l_3
Mass (kg)	m_1	m_2	m_3
l_{c_i} (m)	l_{c_1}	l_{c_2}	l_{c_3}
I_{z_i} (kg.m^2)	$I_{z_1} = p_1$	$I_{z_2} = p_4$	$I_{z_3} = p_5$

3. Model-based Nonlinear Controllers

This section presents three advanced control strategies designed using the full dynamic and kinematic models of the 3 DoF RRR planar manipulator. By leveraging model-based nonlinear control principles, joint space position control, Cartesian space position control, and hybrid position/force control are developed. These controllers aim to improve trajectory tracking accuracy, dynamic compensation, and interaction capabilities, offering high-performance solutions for planar robotic applications.

3.1. Joint Space Position Controller

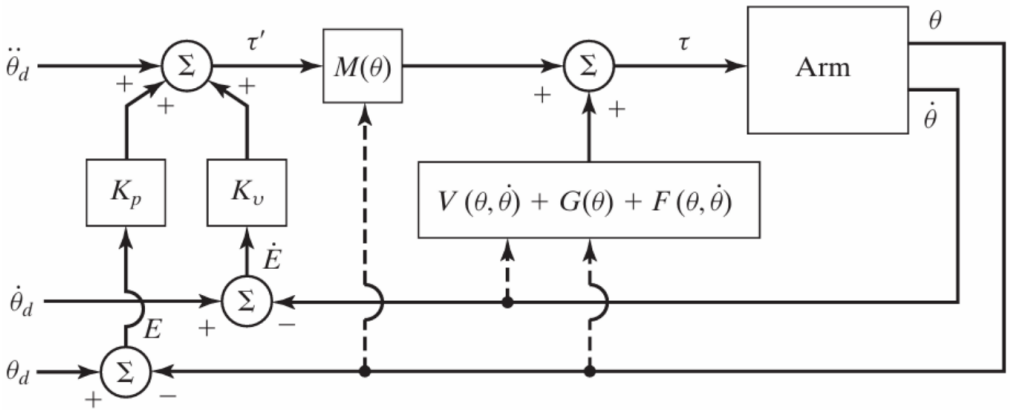


Figure 3: Model-based Joint Space Control Schematic [3]

A model-based nonlinear control law is employed to regulate the joint angles of the 3 DoF RRR planar manipulator in joint space. The control input τ is designed to ensure asymptotic convergence of the joint tracking error by incorporating full dynamic compensation, as expressed in (20):

$$\tau = M(\theta)\ddot{\theta} + C(\theta, \dot{\theta}) + K_p e + K_d \dot{e} \tag{20}$$

The position tracking error vector e is defined as the difference between the desired and actual joint angles (21):

$$e = \theta_d - \theta = \begin{bmatrix} \theta_{1d} - \theta_1 \\ \theta_{2d} - \theta_2 \\ \theta_{3d} - \theta_3 \end{bmatrix} \tag{21}$$

The derivative of the error is formulated as (22), and is used to introduce damping into the closed-loop dynamics:

$$\dot{e} = \dot{\theta}_d - \dot{\theta} \tag{22}$$

To enhance real-time implementation and improve robustness against modeling uncertainties, an alternative expression of the controller is adopted in (23), where the dynamic model is evaluated at the actual joint state q_a:

$$\tau = M(q_a)(\dot{q}_d + K_d\dot{e} + K_p e) + C(q_a, \dot{q}_a)\dot{q} \tag{23}$$

The proportional and derivative gain matrices K_p and K_d are defined as diagonal matrices with positive elements to ensure decoupled joint control (24–25):

$$K_p = diag(k_{p1}, k_{p2}, k_{p3}) \tag{24}$$

$$K_d = diag(k_{d1}, k_{d2}, k_{d3}) \tag{25}$$

The schematic representation of the joint space controller is illustrated in Figure 3, showing the flow of error signals, feedback terms, and the incorporation of the manipulator's dynamic model (Figure 3). The corresponding Simulink implementation is provided in Figure 4, where each control component, including the PD gains and dynamic compensation modules, is embedded into a real-time simulation framework (Figure 4).

The resulting control law enables accurate and stable tracking of joint reference trajectories while compensating for nonlinearities and dynamic couplings inherent in multi-link robotic systems.

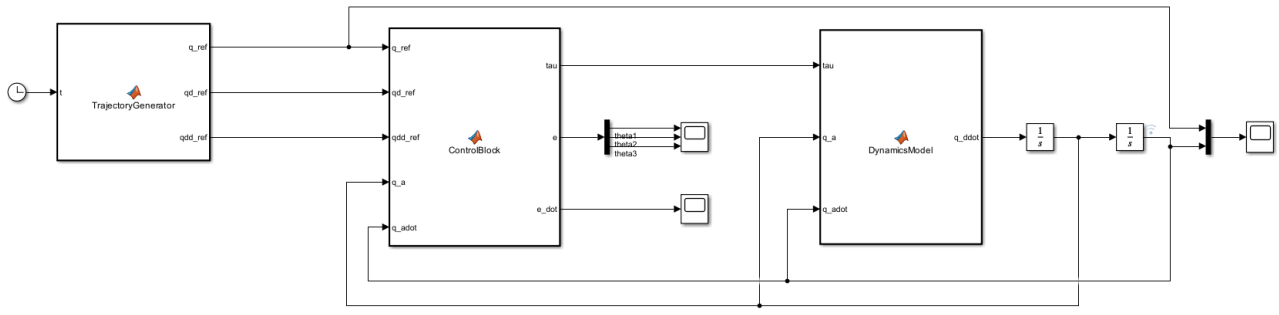


Figure 4: Model-based Joint Space Control in Simulink

3.1.1. Results for Joint Space Position Controller

The performance of the model-based joint space controller is evaluated through simulation under predefined joint reference trajectories. The transient and steady-state responses of the tracking error for all three joints are shown in Figure 5. As observed from the curves, the joint position errors initially

exhibit oscillatory behavior due to the controller's dynamic response to step changes. However, all errors converge to zero asymptotically, demonstrating the effectiveness of the feedback law in ensuring error elimination over time.

Each joint error trajectory converges within a finite settling time, with no residual steady-state error, confirming that the PD terms in conjunction with dynamic compensation successfully eliminate tracking discrepancies. The magnitude and damping characteristics of the transients vary among the joints, which is attributable to differences in inertial and link length parameters across the kinematic chain.

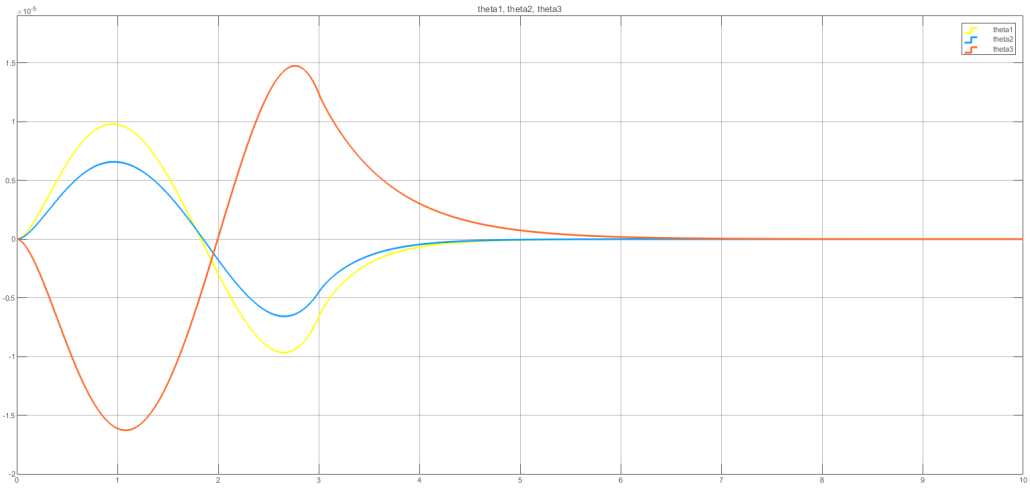


Figure 5: Joint Position Errors for Model-based Foint Space Position Controller

In addition, Figure 6 compares the desired joint trajectories with the actual responses for all three joints. The results show close agreement between the reference and actual trajectories, with minimal overshoot and negligible steady-state deviation. The second and third joints display slightly faster convergence compared to the first, which may be associated with their lower effective inertia and more direct influence of the control torque.

These results validate the proposed control structulhre's ability to compensate for the manipulator's nonlinear dynamics, including joint coupling and velocity-dependent effects, leading to robust trajectory tracking across all degrees of freedom.

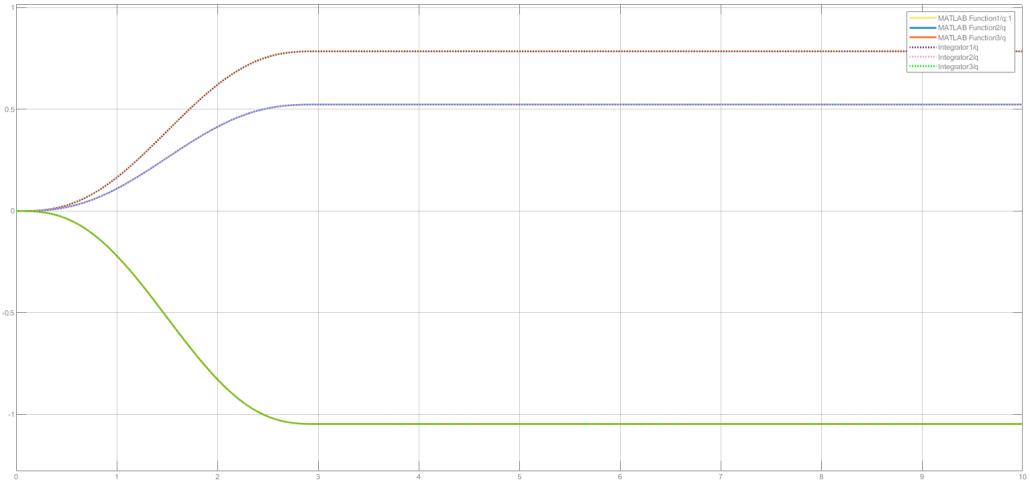


Figure 6: Desired Versus Actual Joint Positions for Model-based Joint Space Position Controller

3.2. Cartesian Space Position Controller

Position control in Cartesian space is a control approach that provides direct access to the desired position and speed values of the end-effector in robotic systems. In this control approach, error signals are defined directly according to the position and speed information of the end-effector in Cartesian coordinates (Figure 7). Position and speed error expressions are as follows:

$$e = X_d - X \tag{26}$$

$$\dot{e} = \dot{X}_d - \dot{X} \tag{27}$$

Here, X_d and $\dot{X}_d$ are the target position and velocity, and X and $\dot{X}$ are the actual end effector position and velocity, respectively. These error values are used together with the proportional (K_p) and derivative (K_v) feedback gains to calculate the control force F acting on the end effector as follows:

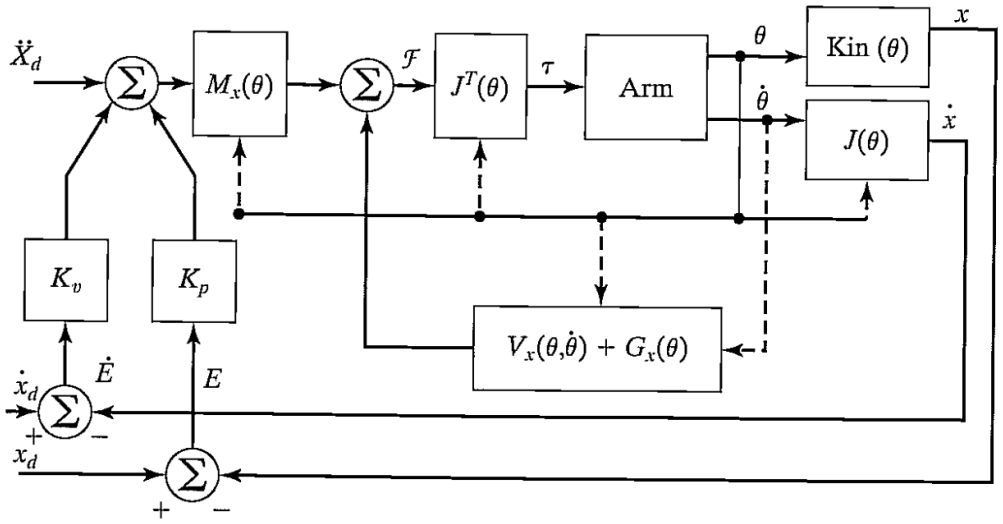


Figure 7: Model-based Cartesian Space Control Schematic [3]

$$F = M_x(\theta)\left(\ddot{X}_d + K_v\dot{e} + K_p e\right) + V_x(\theta,\dot{\theta}) + G_x(\theta) \tag{28}$$

The force calculated in Cartesian space is multiplied by the transpose of the Jacobian matrix to be converted into torque in joint space:

$$\tau = J^T(\theta)F \tag{29}$$

In the Simulink model developed in this study, the following control structure is implemented to perform the Cartesian position tracking of the end effector: First, the Quintic_Trajectory block generates the time-varying position, velocity and acceleration references for the end effector. These references are processed in the CBC (Cartesian Based Controller) block, where the control force F is calculated using the position and velocity errors (e, $\dot{e}$), PD gains (K_p, K_v) and the acceleration reference $\ddot{X}_d$. The calculated force is multiplied by the transpose of the Jacobian matrix to obtain the torque value τ.

This torque is sent to the Dynamic_Model block, where the dynamic model of the manipulator is applied to simulate the behavior of the system. In addition, the Forward_Kinematics block is used to calculate the instantaneous position of the end effector. The Jacobian matrix $J(\theta)$ required for speed transformations and force/torque transformations is generated in real time in the Jacobian block.

The dynamic terms used in the model are transformed into Cartesian space as follows:

$$M_x(\theta) = J^{-T}(\theta)M(\theta)J^{-1}(\theta) \tag{30}$$

$$V_x(\theta,\dot{\theta}) = J^{-T}(\theta)\left[V(\theta,\dot{\theta}) - M(\theta)J^{-1}(\theta)\dot{J}(\theta)\dot{\theta}\right] \tag{31}$$

$$G_x(\theta) = J^{-T}(\theta)G(\theta) \tag{32}$$

Here $V(\theta,\dot{\theta}) = C(\theta,\dot{\theta})\dot{\theta}$ is defined as. If the effect of gravity is neglected, $G(\theta) = 0$ can be taken.

Thanks to this structure (Figure 8), the end effector can follow the reference trajectory with high accuracy; force-based dynamic feedback control provides stable and precise position control by balancing the nonlinear effects in the system.

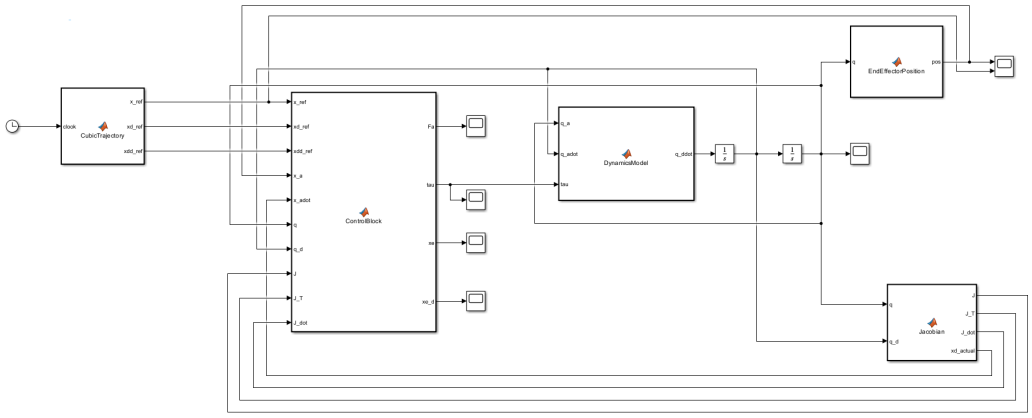


Figure 8: Model-based Cartesian Space Control in Simulink

3.2.1. Results for Cartesian Space Position Controller

The performance of the Cartesian space position controller was evaluated through a simulation in which the end-effector was commanded to follow a circular reference trajectory in the XY plane. The simulation was to assess the controller's ability to perform accurate path tracking using full kinematic and dynamic model-based control in task space. The desired Cartesian trajectory was defined as:

$$x_d(t) = 0.4 + 0.25\cos\left(\frac{2\pi t}{10}\right), \quad y_d(t) = 0.25 + 0.25\sin\left(\frac{2\pi t}{10}\right) \tag{33}$$

which describes a circular path centered at $(0.4, 0.25)$ m with a radius of 0.25 m, completed over a duration of 10 seconds.

The control strategy is based on operational space control. The position and velocity errors are defined as:

$$\mathbf{e} = \mathbf{x}_{ref} - \mathbf{x}_{actual}, \quad \dot{\mathbf{e}} = \dot{\mathbf{x}}_{ref} - \dot{\mathbf{x}}_{actual} \tag{34}$$

The desired Cartesian acceleration is computed using a PD control law:

$$\mathbf{a}_{des} = \ddot{\mathbf{x}}_{ref} + K_v \dot{\mathbf{e}} + K_p \mathbf{e} \tag{35}$$

The control force in Cartesian space is then calculated as:

$$\mathbf{F}_a = M_x \mathbf{a}_{des} + V_x + G_x \tag{36}$$

Here, M and V represent the joint-space inertia and velocity-dependent terms, respectively, while G is the gravity vector, which is neglected due to the planar nature of the system ($G = \mathbf{0}$). The joint torques are finally computed by mapping the Cartesian force to joint space:

$$\boldsymbol{\tau} = J^T \mathbf{F}_a \tag{37}$$

The proportional and derivative gain matrices used in the simulation were:

$$K_p = \text{diag}(100,\ 1000,\ 30), \quad K_v = \text{diag}(100,\ 70,\ 10) \tag{38}$$

These values were selected based on axis-specific dynamics, with larger gains in the Y-direction to accommodate faster response characteristics, while ensuring overall system stability.

As depicted in Figure 9, the end-effector position errors along both X and Y directions rapidly converge to less than 0.1 mm within the first 2 seconds, and remain negligible throughout the remaining trajectory. This demonstrates the controller's high precision and stability.

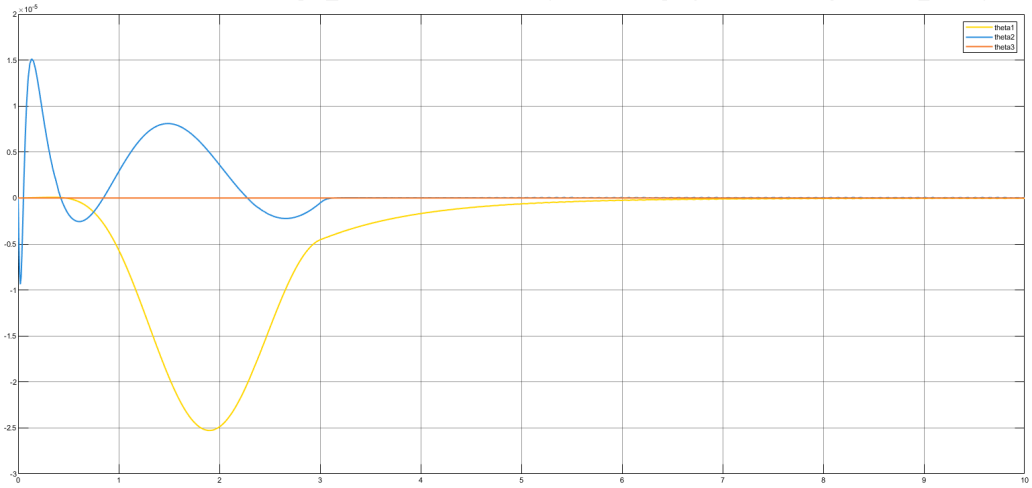


Figure 9: End-effector Position Errors for Model Based Cartesian Space Position Controller

Figure 10 illustrates the actual and desired trajectories of the end-effector in Cartesian space. The two paths are nearly indistinguishable, indicating excellent tracking performance and effective compensation of nonlinear dynamics, inertia coupling, and velocity-dependent effects. The proposed model-based Cartesian controller achieved precise and stable trajectory tracking with minimal overshoot and no steady-state error, confirming the efficacy of full dynamic compensation in task-space motion control for the 3-DoF RRR planar manipulator.

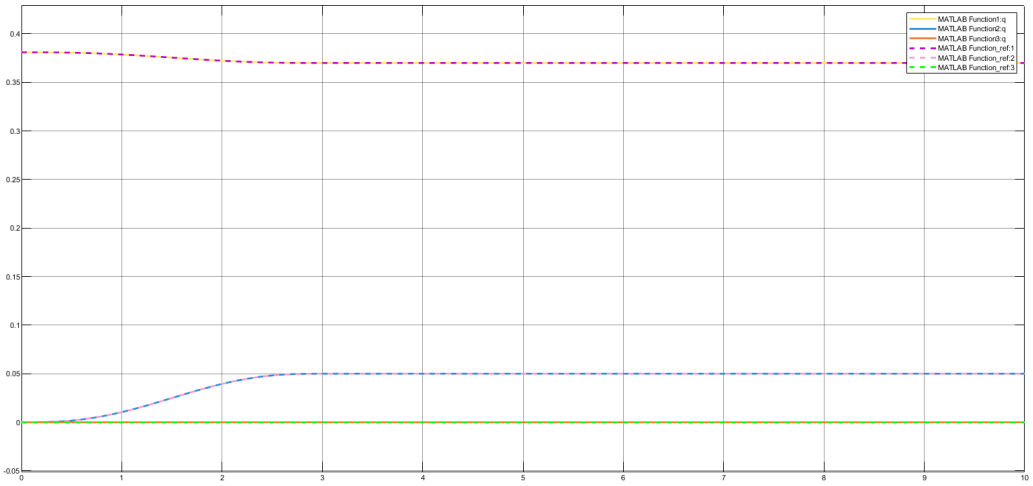


Figure 10: Desired Versus Actual End-effector Positions for Model-based Cartesian Space Position Controller

3.3. Cartesian Space Hybrid Position/Force Control

The hybrid position/force control method in Cartesian space is designed to enhance the physical interaction of the robotic manipulator with its environment, especially through the end-effector. This approach allows the robot to simultaneously control position in certain directions and force in others. It is particularly useful for tasks that involve surface contact such as insertion, assembly, or polishing.

In the proposed method, position control is applied in the x and y directions, while force control is applied in the z direction, which is perpendicular to the 2D plane of motion of the 3 DoF RRR planar robot manipulator. The control system separates the position and force control components and combines them to compute the total Cartesian force.

The overall schematic structure of the hybrid control system is shown in Figure 11, where both control loops operate in parallel. The Simulink implementation of this structure is presented in Figure 12.

The position controller generates the following Cartesian force:

$$F_p = M_x(\theta)\left(\ddot{X}_d + K_v\dot{e} + K_p e\right) \tag{39}$$

In parallel, a force controller regulates the contact force in the z-direction. The contact force with the environment is modeled using a linear spring:

$$f_e = k_e x_z \tag{40}$$

The error between the desired force f_d and the sensed environmental force f_e is:

$$e_f = f_d - f_e \tag{41}$$

Using a damped spring-mass model, the force control output is defined as:

$$F_f = m\left[k_{pf}k_e^{-1}e_f - k_{vf}\dot{x}_z\right] + f_d \tag{42}$$

The selection matrices S_p and S_f determine which directions are subject to position or force control, respectively:

$$S_p = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad S_f = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{43}$$

The total control force in Cartesian space is formed by combining the outputs of the position and force controllers:

$$F = S_p F_p + S_f F_f \tag{44}$$

Finally, the resulting force vector is mapped into joint space as torque input using the transpose of the Jacobian matrix:

$$\tau = J^T(\theta) F \tag{45}$$

This hybrid approach enables the robot to follow the desired trajectory in free directions while simultaneously applying the desired force on the environment in constrained directions, thus achieving compliant and accurate interaction.

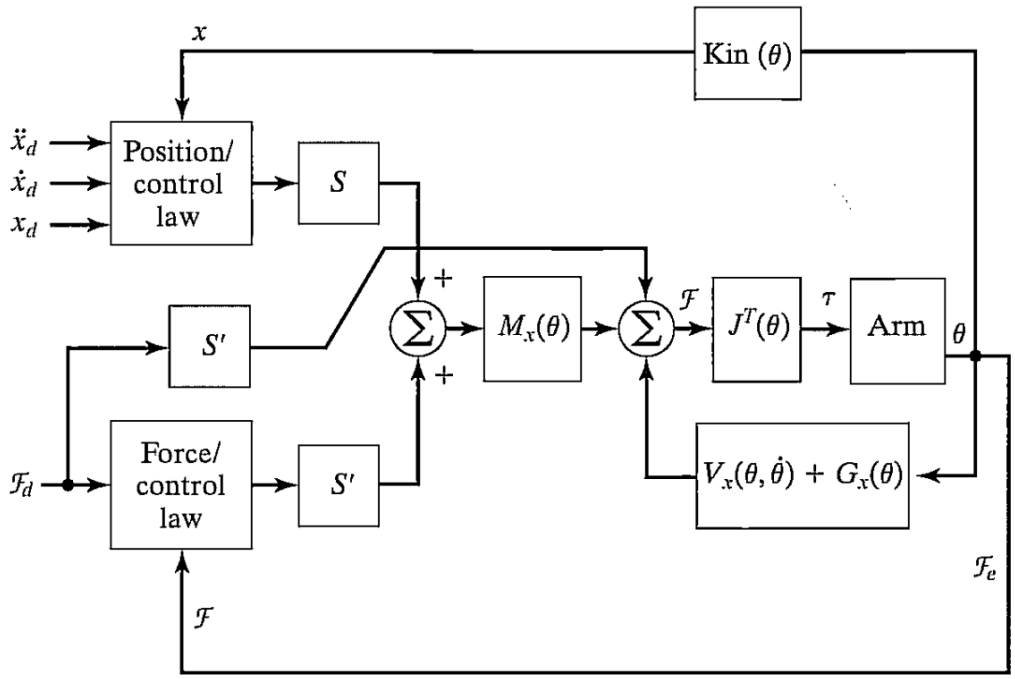


Figure 11: Model-based Cartesian Space Hybrid Position/Force Control Schematic [3]

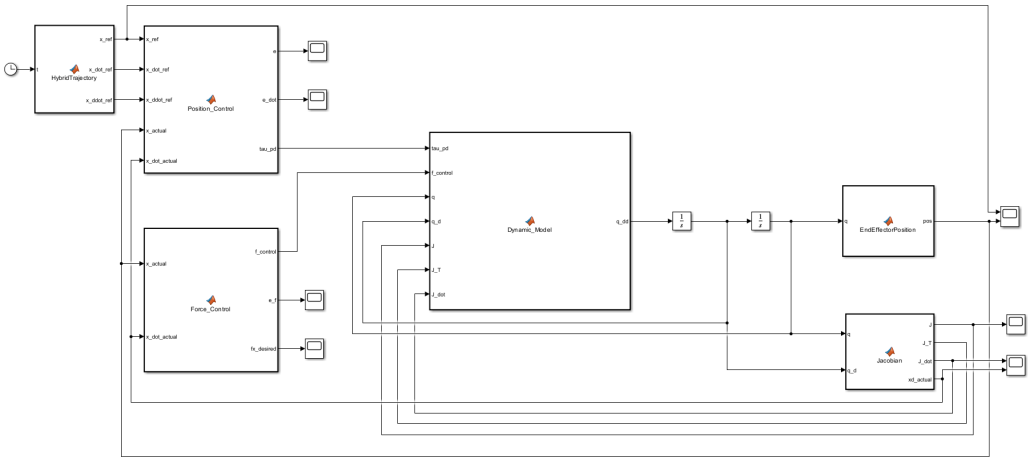


Figure 12: Model-based Cartesian Space Hybrid Position/Force Control in Simulink

3.3.1. Results for Hybrid Position/Force Controller

In this study, the hybrid position/force control strategy was evaluated under a scenario in which **position control was exclusively applied along the Y-axis**, while **force control was applied along the X-axis** to simulate compliant interaction with an environment. The desired end-effector motion in the Y direction followed a sinusoidal trajectory defined as:

$$y_d(t) = 0.4 + 0.2\cos(2\pi t)$$

At the same time, the desired contact force in the X direction was kept constant at **2 N**, simulating interaction with a stiff surface using a linear spring-damper model. The controller used the following parameters:

- Force control gains: $k_{pf} = 5$, $k_{vf} = 2$, $k_e = 50$, $m_{\text{eff}} = 1$
- Position control gains: $k_p = 100$, $k_v = 20$

Figure 13 presents the error profiles for both control components: the **force tracking error in the X direction** and the **position tracking error in the Y direction**. The force control block maintained the X-axis contact force with a high level of precision, keeping the error under 1 mN throughout the simulation. Simultaneously, the Y-axis position controller enabled the end-effector to follow the desired motion profile with negligible steady-state error and fast convergence.

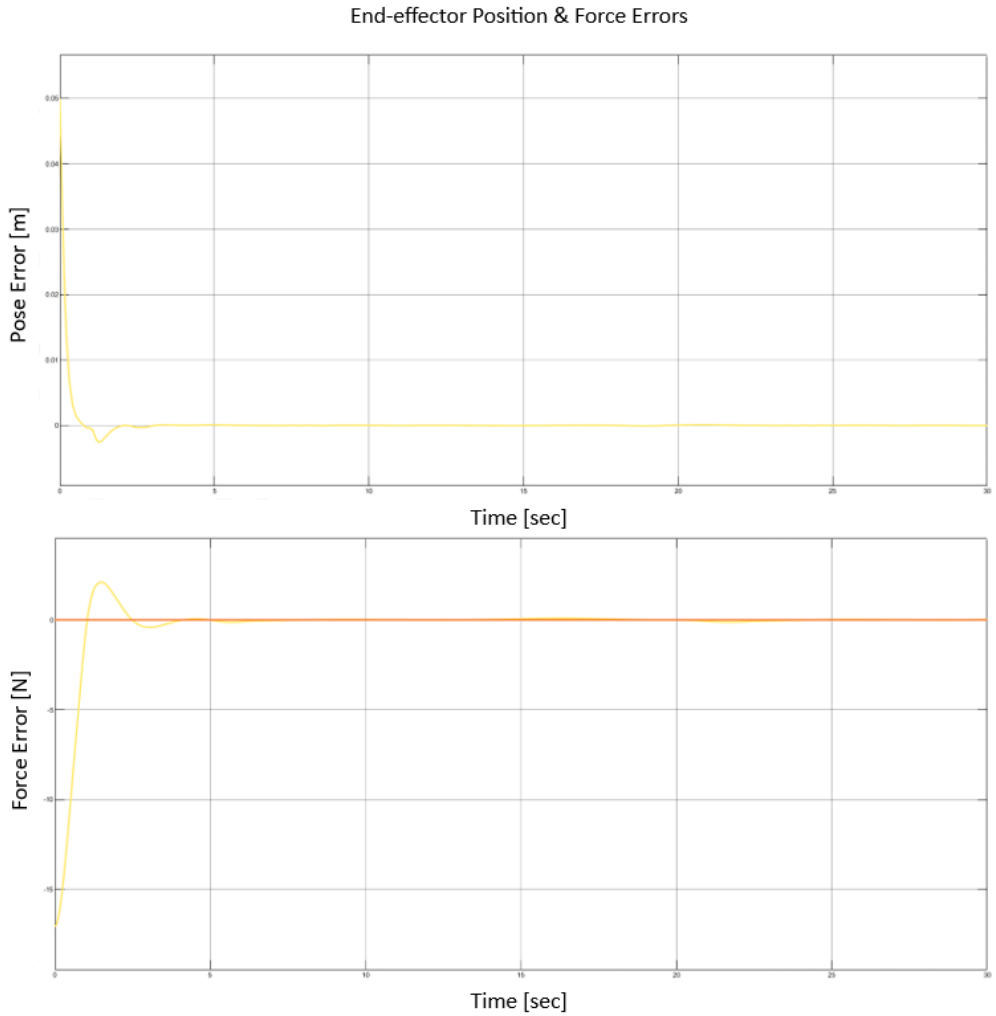


Figure 13: End-effector Position Errors on X and Y for Model-based Hybrid Position/Force Controller

Figure 14 displays the actual and desired trajectories in the Y direction, along with the X-axis force response. The results demonstrate a clear alignment between the desired and measured values, confirming the effectiveness of the decoupled hybrid control strategy. These results verify that the proposed hybrid controller structure, despite applying position and force regulation along orthogonal axes, successfully enables simultaneous motion tracking and environment interaction—making it ideal for planar tasks such as sliding, controlled surface contact, or guided movement with contact force regulation.

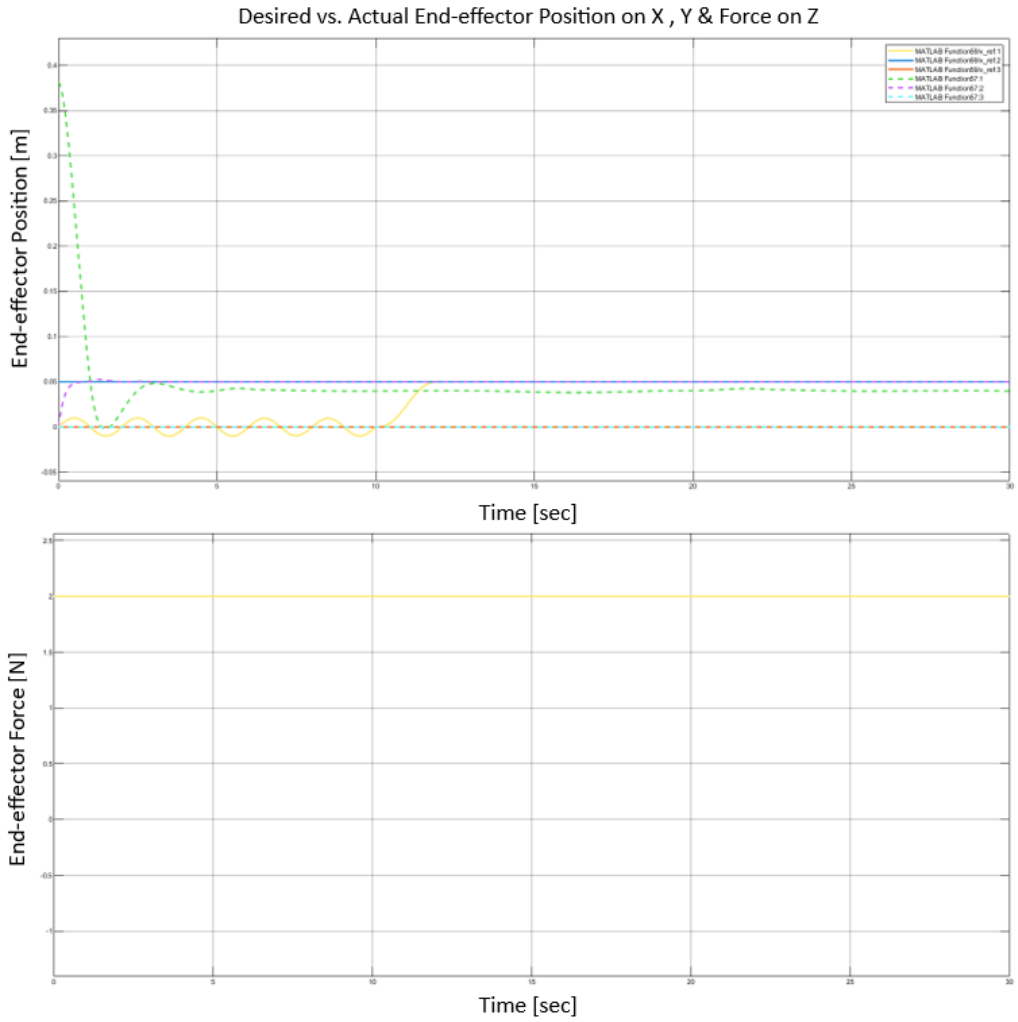


Figure 14: Desired Versus Actual End-effector Positions on X and Y and Force on Z for Model-based Hybrid Position/Force Controller

4. Conclusion

In this study, a model-based nonlinear control framework was developed for a 3-DoF RRR planar robotic manipulator by integrating complete kinematic and dynamic models into three control strategies: joint space position control, Cartesian space position control, and a hybrid position/force control scheme. Each controller was designed and implemented in MATLAB/Simulink and evaluated through simulation-based experiments. The joint space controller achieved accurate trajectory tracking by fully compensating for nonlinear dynamics and joint coupling. All joint position errors converged to zero, demonstrating fast transient response and stable steady-state performance. The Cartesian space controller enabled precise end-effector motion in task space by utilizing Jacobian-based transformations and dynamic compensation. Position errors along the X and Y axes were reduced, validating the controller's effectiveness in managing complex planar trajectories. Hybrid position/force control strategy was introduced in theory to regulate force along the constrained Z-direction. Future work will aim to extend this framework by incorporating real-time force feedback and active interaction control. Overall, the proposed control architecture effectively demonstrates the potential of model-based nonlinear methods in enhancing the precision, stability, and adaptability of planar robotic systems.

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